

# Renlong Zhang

Staff / Senior Software Engineer & Tech Lead · Distributed Backend Systems, Core Foundations & Data Infrastructure  
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## SUMMARY

Staff-level IC software engineer and Tech Lead with 10+ years of experience architecting and scaling mission-critical distributed backend platforms, high-QPS developer APIs, fine-grained access-control systems, and fault-tolerant monetization & data pipelines at **Google**, **YouTube**, **Chainalysis**, **Roblox** and **Huawei**. Currently serving as Tech Lead for **Google Ads BTI / BTO** (a projected **\$2B+/year** post-install & post-open conversion platform on Google Play), having previously served as **Staff Software Engineer & Tech Lead for 9 engineers** delivering **\$10M+/quarter** data platforms and built foundational auth, public APIs, and conversational AI engines on GCP.

## EXPERIENCE

**Senior Software Engineer & Tech Lead** 07/2025 – PRESENT  
*Google · Google Ads (BTI / BTO) & YouTube RevTube*

- **Tech Lead — BTI & BTO (\$2B+/yr projected revenue):** Serve as Tech Lead for **BTI (Beyond the Install)** and **BTO (Beyond the Open)**, flagship ad-serving and conversion platforms projected at **\$2B+ in annual revenue** that orchestrate low-latency, targeted post-install (BTI) and post-open (BTO) conversion workflows (registrations, in-app purchases) at planet scale.
- **End-to-end architecture & launch of BTO on Google Play (Hands-on IC & TL):** Personally leading the end-to-end distributed system architecture, cross-service RPC contracts, hands-on core implementation, and production launch of **BTO on Google Play** under strict latency and reliability SLAs.
- **Technical strategy & architecture governance:** Define multi-quarter technical roadmap and architectural standards across the team; lead system design reviews, production readiness reviews, and code reviews across critical serving and conversion paths.
- **YouTube RevTube — Planet-scale monetization, payouts & tax infrastructure:** Architected and scaled fault-tolerant financial and international tax compliance pipelines (Spanner, FlumeJava) computing creator earnings, enforcing global withholding rules, and guaranteeing zero-discrepancy payouts for millions of YouTube creators and high-volume partners worldwide.

**Staff Software Engineer & Tech Lead** 11/2022 – 07/2025  
*Chainalysis*

- **Staff IC & Tech Lead (9 engineers):** Led system architecture, multi-quarter technical roadmap, and engineering execution for a 9-engineer backend & data infrastructure team owning the company's highest-revenue foundational data platforms.
- **DeFi Metadata Suite (\$10M+/quarter revenue):** Spearheaded end-to-end architecture and hands-on implementation of a distributed blockchain ingestion, indexing, and API/data-cloud serving platform, driving **\$10M+ in quarterly revenue**.
- **High-throughput Price Service:** Architected a horizontally scalable, low-latency backend service ingesting and serving massive real-time and hourly price-data feeds with high availability and strict tail-latency SLAs.
- **Ascriber entity & data-integrity platform:** Designed and built an event-driven storage, validation, and quality-control pipeline guaranteeing cross-chain entity consistency and schema integrity at scale.

**Senior Software Engineer** 11/2018 – 11/2022  
*Google · GCP & YouTube*

- **Fine-grained Access Control & AuthZ (YouTube Live):** Architected and built the foundational access-control and authorization system securing core YouTube Live data entities across complex multi-actor, delegated, and enterprise permission boundaries.
- **Planet-scale Public APIs & Core Services (YouTube Data API v3):** Designed and scaled high-QPS backend services and entity-management RPCs powering creator workflows and the public **YouTube Data API v3** used by millions of developers globally.
- **Conversational AI Backend & Cloud Talent Solution (GCP):** Built and operated a stateful **Conversation Engine** backend powering enterprise conversational chatbots alongside high-scale job-matching and search APIs on Google Cloud Platform.

**Full-Stack Software Engineer** 01/2018 – 11/2018  
*Roblox*

- Designed and shipped high-concurrency social graph ("Play Together" game-card sharing) and real-time mobile chat backend/frontend systems, and integrated platform-wide trust & safety profanity filtering across distributed services.

**Mobile Developer & Tech Lead** 09/2017 – 01/2018  
*Huawei Technologies*

- Served as Tech Lead for a 6-engineer team architecting and shipping cross-platform (Android & iOS) real-time health-monitoring and low-latency camera/wearable streaming systems.

## CORE FOUNDATIONS

Distributed systems & low-latency RPC · High-QPS developer & product APIs · Fine-grained AuthZ & access control · Fault-tolerant billing & data pipelines · Event-driven async systems (Pub/Sub, queues) · Staff technical leadership & roadmaps

## INFRASTRUCTURE & STORAGE

Spanner, distributed SQL/NoSQL · Flume (FlumeJava), streaming & batch · Google Cloud Platform (GCP), gRPC / Protobuf

## LANGUAGES

C++, Java (Google readability-certified) · Python, TypeScript (expert) · SQL

## APPLIED DOMAINS

Ad-serving & conversion platforms (\$2B+/yr) · Monetization, payments & tax compliance · Conversational AI & public API platforms · Trust, safety & multi-tenant security